

INVESTIGATING THE ROLE OF COMBUSTION CHAMBER LENGTH AND FUEL INJECTION IN PULSE DETONATION ENGINE PERFORMANCE

Report

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CERTIFICATE

I hereby certify that the work which is being presented in the capstone entitled “**Investigating the Role of Combustion Chamber Length and Fuel Injection in Pulse Detonation Engine Performance**” in partial fulfillment of the requirement for the award of degree of **Bachelor of Technology** and submitted in Department of Aerospace Engineering, Lovely Professional University, Punjab is an authentic record of my own work carried out during period of Capstone under the supervision of **Dr. Bilji C Mathew , Associate Professor**, Department of Aerospace Engineering, Lovely Professional University, Punjab.

The matter presented in this capstone has not been submitted by me anywhere for the award of any other degree or to any other institute.

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ABSTRACT

This document offers a thorough comparative examination of two advanced detonation-based propulsion systems: Pulse Detonation Engines (PDEs) and Rotating Detonation Engines (RDEs). Both systems utilize the principles of supersonic combustion to achieve improved efficiency, thrust, and specific impulse relative to traditional engines. The analysis delves into the operational mechanisms, performance indicators, and challenges associated with PDEs and RDEs, underscoring their benefits in terms of thermodynamic efficiency, fuel adaptability, and suitability for aerospace propulsion. While PDEs are characterized by their straightforward design, they encounter issues such as mechanical stress, unstable detonation cycles, and elevated noise levels. Conversely, RDEs facilitate continuous detonation, which enhances operational stability and allows for more compact designs, yet they grapple with challenges related to wave control and material durability. The document further investigates recent progress in detonation dynamics, structural materials, and integration methods for real-world aerospace applications. This comparative study not only identifies specific applications and technological hurdles but also establishes a foundation for future research and the potential implementation of detonation-based propulsion systems in aerospace endeavors.

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DECLARATION

We, the undersigned members of the Team, hereby affirm our dedication to excellence, innovation, and collaboration in the advancement of propulsion technology. With steadfast commitment, we vow to expand the horizons of aerospace engineering, lead the charge in PDE research, and enhance the development of global aerospace capabilities.

Leveraging our combined expertise, relentless efforts, and unified vision, we strive to propel the project towards groundbreaking innovations within the industry. Our objective is to challenge established norms, foster creativity, and redefine the potential of detonation-based propulsion for the advancement of aerospace technology and the betterment of humanity.

With unwavering determination to tackle challenges, embrace innovation, and maintain the highest standards of professionalism, we embark on this endeavor with enthusiasm, integrity, and a collective aim of shaping the future of the aerospace sector.

Signed,

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Table of Contents

S.no	CONTENT	Page no
1.	Abstract	iv
2.	Introduction	1
3.	Literature Review	2
4.	Research Gap	11
5.	Objectives of The Study	12
6.	Methodology	13
7.	Experimental Setup	14
8.	Results	23
9.	Conclusion	
10.	References	

List of Figures

1	Schematic Diagram Of Pulse Detonation Engine	2
2	Theoretical Dependence of Specific Impulse on different propulsion systems and different fuels[11]	3
3	Operational stages of PDE	4
4	Experimental set up of PDE[19]	9
5	Working Model of Pulse Detonation Engine	12
6	CAD Section View of Shchelkin Spiral and Detonation Chamber	12
7	Image of PDE In Action	13
8	Image of Fuel injector Control Circuit using Pulse Modulator	14
9	Fuel Injectors with Y Section	14
10	Buffer Tank	15
11	Solenoid Valve	15
12	Spark Plug	15
13	Schematic Diagram of Injector Control Circuit	16
14	NI DAQ	21
15	S-type load cell	21

List of Abbreviations

ABBREVIATED FORM	EXPANDED FORM
PDE	Pulse Detonation Engine
RDE	Rotational Detonation Engine
DDT	Deflagration-to-Detonation Transition
CD	Converging-Diverging
PDRE	Pulse Detonation Rocket Engines
MHD	Magnetohydrodynamic
TOGW	Take Off Gross Weight

2. Introduction

The Pulse Detonation Engine is a propulsion system that has garnered a lot of attention in the past ten years because of its many benefits over conventional jet engines. A PDE works through an entirely different operational method as compared to standard jet turbines. PDE technology relies on detonation as its power source which performs supersonic combustion more efficiently than traditional engines pursue continuous combustion. The main operating principle of PDEs involves controlling detonations in a combustion chamber which produce quick bursts of controlled explosions generation of thrust. The high-pressure shock waves created by detonations

A PDE functions through four sequential steps starting with chamber filling by fuel-air mixture followed by mixture ignition and the propagation of the detonation wave before exhausting the gases. High-frequency operation allows this cycle to repeat until thrust is created through the device. The unique operation of PDEs produces engine noise that releases rapidly exploding sounds resulting in the detonation engine name.

PDEs demonstrate exciting features because of their capability to achieve improved thermodynamic performance. The detonative combustion process used in PDEs achieves better fuel consumption than traditional deflagration systems thus leading to reduced fuel needs and better thrust efficiency as well as reduced operating expenses. The characteristics that draw defense agencies to PDEs apply to military aircraft applications and high-speed missiles together with space launch systems

The basic design structure of PDE serves as an additional advantage. Because the PDE operates without turbojet or turbofan engine moving elements it provides fewer maintenance requirements while potentially achieving reduced weight. The development of fully operational PDEs faces multiple problems on their path to completion. Scientists continue active research to determine how they can control stable detonation and handle extreme heat and reduce high noise production.

Test experiments from the last two decades yielded positive results while researchers explore Pulse Detonation Turbopumps as hybrid engine solutions. Aircraft systems can accept this hybrid propulsion system by uniting PDE detonation-based combustion with standard turbomachinery.

The Pulse Detonation Engine brings forth substantial advancements in propulsion systems field. The engine uses detonation power combined with engineering progress to extend beyond conventional engine capabilities. PDE-based research and development continues due to its potential for improved efficiency and performance although the technology endpoints are currently evolving.

3. Literature Review

Fundamental Principles of Pulse Detonation Engines

Pulse Detonation Engines represent a new novel form of propulsion that offers the potential for high combustion efficiency and performance with reduced hardware complexity compared to the current rocket-based, turbines, and ramjet/scramjet engines[9]. The Pulse Detonation Engine concept utilizes a repetitive steady cycle of detonation wave initiation, propagation, and expulsion to produce thrust[10]. Detonation is a supersonic combustion process in which the unburned gas's reaction front moves faster than sound. A shockwave is produced, and it spreads throughout the mixture. As a result, chemical energy is converted to thrust quite effectively. PDEs offer the potential for high performance because the rapid detonation process results in constant volume combustion with very high operating frequencies and combustion chamber pressures [9]. PDEs are predicted to be very efficient and offer good thrust characteristics from low subsonic to high supersonic flight regimes. PDEs also have the potential to operate at very high energy densities, allowing simple and compact combustion chambers to be used [9].

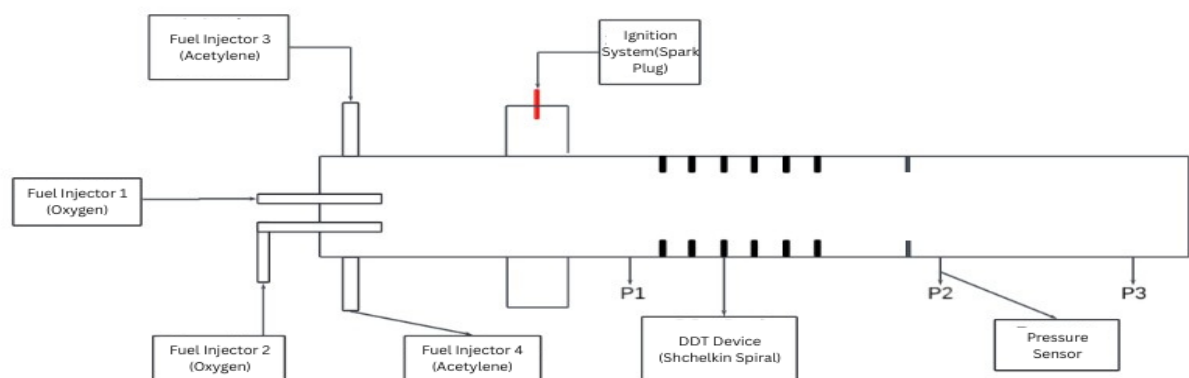


Figure 1 Schematic Diagram Of Pulse Detonation Engine

Detonation vs Deflagration

Detonation, a supersonic combustion process, has a reaction front in fuel-air mixture velocities that are normally greater than 1.8 km/s accompanied by more than tenfold pressure rise[11]. Detonation velocities are up to 3 km/s when using oxygen-enriched mixtures and pressure rises are even higher[11]. Traditionally, detonation has been associated with accidental explosions; they have an impact on pressure transients, which are quite high. Although detonative combustion has been around for some time, its full practical use has been restricted especially due to the recent developments in detonation

engines[11]. However, deflagration produces lower flame velocities that vary between several meters per second and thus require stoichiometric combustion for better-burning velocities[9],[11]. These characteristics make detonation combustion ideal for jet propulsion, enhancing engine performance and allowing for streamlined designs[9],[10],[11]. If detonation combustion is applied to a jet engine, the efficiency of the engine cycle can be theoretically increased even more than 15%[11]. This is because, during detonation, the specific volume of the reacting mixture decreases, making the theoretical efficiency of the cycle even higher than that of constant volume combustion [11].

The specific impulse for such propulsion systems with detonation combustion is also much higher than for conventional engines (Fig.2)[11].

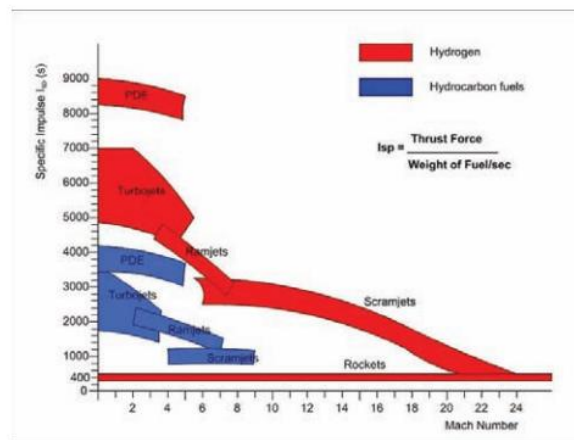


Figure 2 Theoretical Dependence of Specific Impulse on different propulsion systems and different fuels[11]

Operation of a PDE

Like other types of developmental models, the development and operation of PDEs is a systematic process that includes a clear and defined cycle of phases concerning the overall propulsion mechanism[13]. These cycles are fine-tuned to get the absolute best circumstances in which detonation will occur, maximum efficiency of exhaust, and quickest cycling in the least possible time with a minimum buildup of combustion byproducts that might hamper performance[9],[12]. Thanks to the improvement of these stages, PDEs can work at high frequency to provide nearly continuous thrust that can be used in high-speed vehicles and spacecraft. In the next sections, each phase of the PDE cycle is described in detail considering the requirements for the fuel-oxidizer mixture preparation, ignition and detonation, product exhaust, and the final purge step that resets the chamber for subsequent cycles[13]. All the stages in the operation of a Pulse Detonation Engine are shown in the figure given below[14].

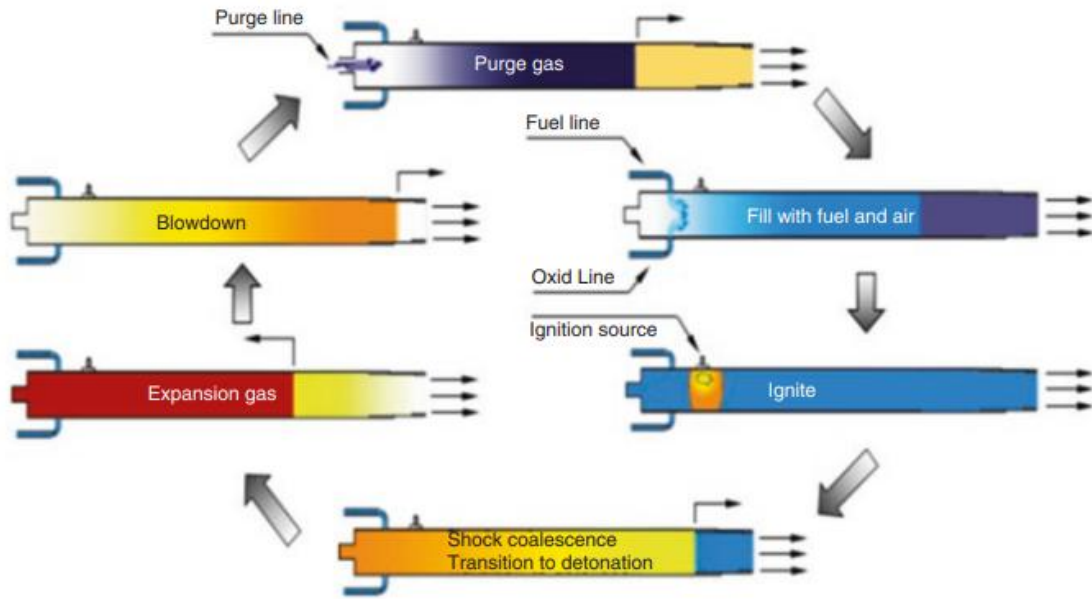


Figure 3 Operational stages of PDE

Filling Stage

This step involves introducing into the detonation chamber of the PDE a fuel-oxidizer combination of hydrocarbon and oxygen or air[12]. The fuel and the oxidizer are well mixed or in other words, emulsified-which is one of the critical steps in detonation[13],[14],[15].

Ignition Stage

When the detonation chamber has a well-mixed fuel-oxidizer combination, the following step is required to induce detonation[12], and this is to ignite the mixture well. In catalytic PDEs, a spark plug is usually employed to provide the ignition energy required mainly to initiate the reaction. In terms of the spark or other ignition source, it means that it transfers enough energy to sustain a small volume of fuel-oxidizer mixture. This local ignition starts with steady burning or deflagration in which the flame front moves at subsonic velocities[13]. As the flame extends, it inflicts heat and pressure on the enclosed gases hence providing a high-energy area in the detonation chamber[14].

Detonation Stage (Combustion).

As the flame propagates through the mixture, the unburnt fuel-oxidizer mixture ahead of it gets compressed[14]. This compression raises the temperature and pressure of the unburnt gases. If conditions are right (i.e., sufficient pressure, temperature, and mixture homogeneity), the flame front accelerates, and a shockwave forms. The combustion then transitions to detonation. This stage is called the **Deflagration-to-Detonation Transition (DDT)**[9],[10],[13],[15]. The DDT process consists of two independent phases: flame acceleration and detonation onset.[15] Both phases are random, making it difficult to

quantitatively predict the transition distance or transition time[15]. The geometry of the detonation chamber can influence the ignition and DDT. For example, a long, straight chamber with a small cross-section can help accelerate the deflagration to detonation.

Expansion and Exhaust Stage

The gases that are beyond the detonation wave are highly heated and are under very high pressures when compared with the pressure for a deflagration wave. Because of the sharply defined pressure differences between the combustion gases and the pressure at the engine surface or out of doors, these gases expand vigorously and attempt to reach the pressure level of the exterior environment. This expansion results in the gas being forced towards the exhaust nozzle[14]. Most of the time a convergent-divergent nozzle is employed at the exhaust section. At the convergent section of the nozzle, the high-pressure gases are compressed in the sense that they are accelerated. Following the throat, the gases expand in the following divergent section where the velocity of the flow rises again to supersonic values[11],[14]. This supersonic exhaust is critical in the generation of higher thrust forces in many PDEs. According to Newton's Third Law of Motion, the forceful ejection of gases out of the nozzle creates a force to force the engine in the direction of travel[10].

Purge Stage

Following each detonation, some of the burning products of combustion CO₂, water vapor, or other chemical residues and the unburnt fuel may remain in the chamber. They can lead to incomplete combustion, detonation efficiency, or even create instabilities in the engines[14]. The detonation process creates highly elevated temperatures inside that chamber depending on the detonating materials. The purge step is helpful to ensure the chamber cools before the next injection of fuels and oxidizers[14]. To expel the product mixture from the detonation chamber, air or an inert gas (like nitrogen) is blown into the chamber[14]. In many cases, air is used for purging because it is readily available, and its introduction may serve as cooling as well.

Repeat Cycle

The whole process above is considered as one cycle. The entire process gets repeated in quick succession to provide constant thrust required for flight. The cycle frequency of a PDE relates to the number of complete rounds of detonation from injection, ignition, expansion, and purging within a second. Cycle frequencies in PDEs tend to vary within a scale of tens to hundreds of cycles per second. For example, a standalone PDE working at 100 would perform one detonation cycle per second and is referred to as a 100 Hz PDE. As PDEs discharge discrete pulses of thrust with each detonation, due to high cycle frequency this thrust appears continuous. Yet, for high cycle frequencies, the basic phases or steps in each detonation cycle need to be properly phased.

Propulsion System Integration of PDE

PDE as a replacement to turbo-ramjet engines

Boeing's Kaemming compares the performance of a pulsed detonation engine (PDE) to a conventional turbo-ramjet engine[16], with a focus on high-speed, supersonic tactical aircraft applications. The project, directed by NASA Langley, was carried out in stages to establish the feasibility and benefits of PDE for different vehicles. The investigation eventually focused on a supersonic tactical aircraft, assessing the PDE's potential in terms of specific fuel consumption (SFC), thrust, weight, and overall vehicle cost.[16]

Key Findings:

Performance: PDEs outperform turbo-ramjet engines in terms of fuel efficiency at high speeds due to lesser entropy generation and increased combustion pressure. However, they have reduced thrust at low speeds because they rely on inlet ram air for pressure, which is not available at lower speeds.[16]

Design implications: The PDE enables flexible engine integration, which can improve the aircraft's aerodynamic profile and reduce supersonic drag. When engines are placed around the trailing edge, drag is reduced, improving high-speed performance.[16]

Weight and Cost: The PDE-powered aircraft had a reduced takeoff gross weight (TOGW) of 11-21% (depending on mission profile) due to its efficient high-speed fuel utilization and lightweight structure. This reduction in TOGW corresponds to an estimated 4% decrease in life cycle costs, mostly due to lower acquisition and operations expenses.[16]

Modular Air Breathing Pulse Detonation engine

Lu et al. [17] created a large-scale, modular PDE prototype that incorporates advancements in ignition and fuel handling technologies meant to interface with broader aerospace applications.

This prototype featured several essential subsystems that meet the operational requirements of PDEs in integrated propulsion systems. Notable advancements included dual rotary valve mechanisms for accurate timing of fuel-air mixture injection and purging cycles, which are vital for sustaining the frequency of the PDE cycle while preserving combustion efficiency[17]. Moreover, the prototype utilized high-energy ignition with plasma assistance to expedite the DDT process, which is crucial for high-frequency operation and aids in reducing the considerable ignition energy generally needed for liquid fuels. To cope with the substantial thermal load produced by continuous detonations, the engine was equipped with pressurized water-cooling channels throughout key components, guaranteeing thermal stability and preventing auto-ignition from residual heat during rapid cycling. In addition, the incorporation of a remote-controlled data acquisition system in this design facilitates real-time monitoring and diagnostics, conforming to aerospace standards for propulsion safety and operational oversight.[17]

The incorporation of Pulse Detonation Engines (PDEs) into propulsion systems offers considerable potential for enhancing efficiency; however, it necessitates meticulous attention to design and operational parameters to maximize performance. Recent

experimental findings indicate that thrust and specific impulse are significantly influenced by variables such as exhaust pipe length, spark plug frequency, and oxidizer pressure[18]. For example, optimal thrust values reaching 28.71 N were recorded with a 400 mm exhaust pipe length using hydrogen/air mixtures at frequencies of 100 Hz, 150 Hz, and 350 Hz. Conversely, detonation stability was compromised at oxidizer pressures exceeding 6 bar due to insufficient fuel penetration into the vortex barrier, highlighting the critical role of aerodynamic configuration in the mixing of fuel and oxidizer as well as in detonation initiation.[18] Furthermore, experiments utilizing hydrogen/oxygen mixtures demonstrated that while detonation was more easily achieved, it resulted in lower time-averaged thrust, illustrating the inherent trade-offs associated with different oxidizer mixtures in propulsion applications. These results emphasize the necessity for continued research into the scaling of PDEs and the formulation of integration strategies that guarantee stable and efficient operation across diverse flight conditions.

Practical challenges in real world applications

While detonation-based propulsion systems, like Pulse Detonation Engines (PDEs) and Rotating Detonation Engines (RDEs), will revolutionize aerospace propulsion, as they can achieve very high efficiencies with minimal fuel consumption, several practical issues remain before such propulsion systems become feasible for practical implementation. The practical issues involve stability, mechanical stress, fuel management, noise and vibration, and integration with current aerospace systems [1] [2] [3].

Stability of Detonation: The major problem in detonation-based engines is the sustainment of a stable and repetitive detonation wave. In a PDE, since every combustion cycle initiates a new detonation, instability could arise from incomplete combustion, flame quenching, or improper synchronization between the fuel injection and ignition systems. If detonation does not fully propagate throughout the fuel-air mixture, loss in thrust may sharply decrease along with decreased efficiency in the engine and increased wear on its components [4] [5].

Maintaining wave coherence and controlled wave behaviour remains at the top of the agenda for detonation research because the instability in wave propagation frequently leads to less-than-expected performance results. Advanced computational modelling and experimentation are necessary to better understand and predict detonation wave behaviour under changing operating conditions [8].

Mechanical and Heat Stress: Detonation-based engines operate at extreme thermal and mechanical stresses, much larger than those experienced in conventional propulsion systems. A detonation process produces a sharp, high-pressure shockwave and cyclically impacts the material of the engine with an associated possibility of fatigue failure over time [2]. In the PDEs, parts of the engine have to endure repeated thermal cycling induced by each combustion event which may result in rapid materials degradation, especially in areas such as the combustion chamber and exhaust nozzle [4].

Great effort is required to design new, lightweight, and heat-resistant materials that will be able to withstand extreme conditions [3]. The present research on ceramic composites, metal matrix alloys, and advanced thermal barrier coatings is directed toward the generation

of materials that can be able to endure these stresses without degrading performance or adding excessive weight to the propulsion system [5].

Fuel Injection and Management: The detonation-based engines are very sensitive to the proper injection of fuel and oxidizer. Both PDEs and RDEs need very finely adjusted fuel delivery systems so that the fuel-air mixture falls within the optimal range for detonation. Deviations in the fuel-to-oxidizer ratio can lead to incomplete combustion, inefficient detonation, or even failure to initiate a detonation wave [7].

In PDEs, fuel has to be injected into the combustion chamber at a high speed, and the time must be adequate enough to mix with the incoming air before detonation is initiated. Timing is important, as delays in fuel injection can cause the detonation to miss the optimal point in the cycle, which results in loss of efficiency.

Emphasis nowadays is placed on more accurate injection technologies, namely pulsed injectors for PDEs and swirl injectors for RDEs, in regard to dynamic conditions for improved fuel-air mixture delivery. Further research is being targeted to optimize fuel injector placement and design to ensure better mixing and reduce turbulence within the combustion chamber [8].

Vibration and Noise: Detonation is a very fast, explosive process, producing tremendous vibration and noise, much more than that produced by conventional combustion engines. For PDEs, each detonation event creates a shockwave that reverberates through the engine structure, causing high levels of vibration. These vibrations pose significant challenges for the structural integrity of the engine and the airframe of the vehicle in which it is installed. Such vibrations may eventually lead to structural fatigue, which might result in catastrophic failure, since it occurs several times due to the length of time involved [6].

Noise mitigation and vibration damping technologies are very crucial in reducing the impact effects of environmental effects and for safety and durability for detonation engines. Studies into acoustic liners, vibration isolation systems, and possible active noise control applications to alleviate vibration caused by detonation effects are ongoing [7].

Integration and Scalability: Perhaps the biggest engineering challenge is how to integrate PDEs and RDEs with current aerospace platforms. Detonation engines are quite dissimilar from traditional propulsion systems, and integration with the current airframe designs and control systems is a challenge [4]. For instance, the significant levels of vibration and noise generated by detonation engines may interfere with the avionics and structural components that are sensitive to such conditions, and hence aircraft designs will need to be changed considerably [8].

Scaling detonation engines for commercial or military use is a very difficult task. While small-scale laboratory models are promising, transitioning them into full-scale engines for practical applications presents additional challenges [2]. For instance, stability in detonation increases the challenge with an increase in the size of the engine, and cooling requirements increase exponentially with larger combustion chambers [6].

Besides the aforementioned technical problems, detonation engines raise safety concerns if applied in real conditions. Detonation combustion is energetic[3]. Therefore, minor faults

in the management of fuel or control of an engine might lead to disastrous failure with a high threat to a vehicle and its crew [7].

Current trends and future research directions Of Pulse Detonation Engines

PDE Fuel Optimization using Ionized Seeds

Experiments involving the addition of Ionized Seeds into pulse detonation have been conducted[19]. Ionized seeds, often salts like potassium carbonate (K_2CO_3), are introduced to a combustion system to improve the electrical conductivity of the gas produced after detonation. These seeds are thermally ionized in high-temperature situations such as in pulse detonation engines, resulting in the release of charged particles. This increases the ionization level of the combustion gases, which can have a variety of benefits, including magnetohydrodynamic (MHD) control and improved detonation stability. A magnetic field and electric current can decelerate or accelerate an ionized gas flow, producing electrical power or turning it into kinetic energy. Magneto-Hydro-Dynamics (MHD) can improve system performance, expand operational range, and enable multi modal missions [20].

Research conducted at the *Nanjing University of Science and Technology*[19] has also revealed that Adding ionized seeds increases the **gas conductivity** during detonation, enhancing the stability of the detonation wave. The effect of ignition frequency was also observed. At 5Hz, the maximum peak pressure was achieved, and stable detonation was maintained. An ignition frequency of 10Hz also yielded a stable detonation, with a slightly lower peak pressure. At 20 Hz, the detonation wave degenerated into a shock wave due to incomplete fuel filling, highlighting that excessively high ignition frequencies affect detonation stability.

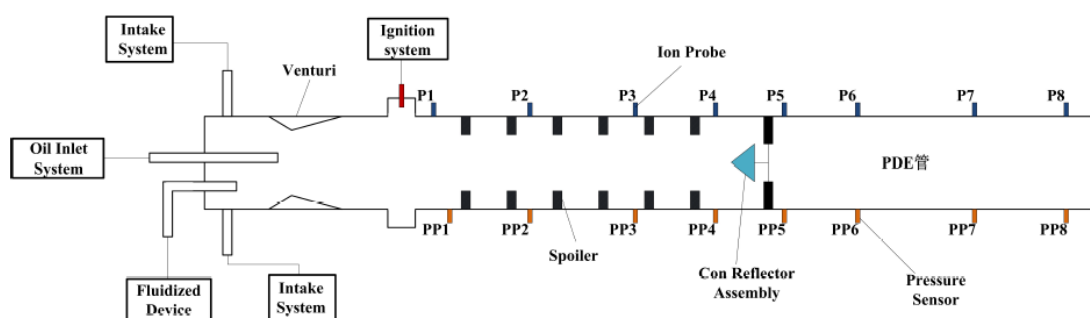


Figure 4. Experimental set up of PDE[19]

PDE Fuel optimization using Biofuels.

Biogas is one of the most promising alternative energy sources and research regarding its use in Pulse detonation engines have been conducted at Universiti Teknologi Malaysia[21]. Their research investigated how the addition of hydrogen to biogas affected detonation

characteristics. The PDE was tested at 10 Hz with a mixture of synthetic biogas (60% CH₄ + 40% CO₂) and oxygen as an oxidizer. Different amounts of hydrogen were added (5%, 10%, 15%, and 20%) to determine the impact on detonation characteristics.

Increasing the methane concentration in the biogas raised the detonation pressure, temperature, and Mach number. However, the presence of CO₂ diluted the fuel, decreasing combustion performance by reducing flame speed and detonation strength.[21] 15% hydrogen addition yielded the optimal detonation performance, increasing detonation pressure by 23% compared to regular biogas. Biogas without hydrogen or with minimal hydrogen content (0%-10%) resulted in deflagration, not detonation.

In conclusion, **Hydrogen-enriched biogas** effectively improves PDE performance by increasing detonation pressure and speed, and the best results were observed with 15% hydrogen, which provided optimal detonation properties without significant trade-offs.[21]

Effect Of Operating Frequency And Ignition Energy

Operating Frequency Effects: Research indicates that as the operating frequency of the engine increases from 20 Hz to 60 Hz[22], the propulsion effect produced by a single cycle of the engine deteriorates. This suggests that there is a threshold frequency beyond which performance declines, with the optimal frequency identified as not exceeding 50 Hz under the tested conditions[22]. Similar research has proved that specific fuel consumption decreases with higher operational frequencies[23]. This is a significant finding as it implies that the engine becomes more fuel-efficient, which is crucial for practical applications[23].

Ignition Energy Influence: A critical finding of the research is that ignition energy significantly affects the early flame acceleration during the detonation process[22]. The results show that increasing the ignition energy from 25 J to 400 J leads to an approximate 18.3% increase in specific impulse[22]. This highlights the importance of ignition energy in enhancing the performance of the Multiphase-Pulse-Detonation Engine.

Challenges and Limitations

Noise and Vibration

Challenge: - Explosions generate pressure waves of exceptionally high amplitude, leading to significant noise and vibrations in structures. This results in acoustic fatigue, harm to engine mounts, and may surpass the acceptable limits for human or equipment tolerance. Consequently, it restricts the use of PDEs in crewed aircraft unless sophisticated damping or insulation technologies are created.

Structural and thermal stress

Challenges :- The frequent high-pressure detonations result in significant thermal and mechanical stress on engine components. The materials utilized must endure cyclic thermal loads and stresses induced by shock. The walls of the tubes and the DDT sections are subject to erosion and fatigue, necessitating the use of materials that possess high strength and can withstand elevated temperatures. Effective thermal management emerges

as a critical concern, particularly in the context of continuous or high-frequency pulse detonation engines.

Lack of Practical Demonstrations

Challenges :- There are limited instances of full-scale, long-duration PDE flights or systems, with the majority of PDE research remaining restricted to laboratory-scale or ground-based evaluations. The shift from theoretical concepts and prototypes to fully operational aerospace systems necessitates a multidisciplinary engineering approach and incurs significant expenses.

4. Research Gap

The development of Pulse Detonation Engines (PDEs) faces several unresolved research challenges that limit their practical use. A key issue is achieving a reliable and consistent Deflagration-to-Detonation Transition (DDT). Although methods like Shchelkin spirals and orifice plates help initiate DDT, a universally optimized design for a quick and effective transition across various conditions is still lacking, affecting engine stability and performance. Additionally, high-frequency operation remains a challenge, as synchronizing ignition, purging, and fuel-air injection at over 100 Hz requires advanced, lightweight valving and timing systems that have not yet been realized. The mixing of fuel and air, especially with liquid fuels like kerosene or JP-10, is crucial; effective vaporization and uniform mixing within milliseconds are essential for successful detonation, but current systems often fail to achieve this, leading to misfires or inefficient combustion. Ignition systems must also function reliably under extreme conditions, yet traditional spark-based systems quickly degrade at high temperatures and pressures. There is a lack of robust, miniaturized ignition alternatives, such as laser or plasma systems, that can sustain long-term PDE operation. Material limitations pose another significant research gap, as the rapid thermal and pressure changes during detonation lead to fatigue, erosion, and failure, but long-term studies on material degradation in PDE conditions are scarce. Furthermore, PDE cooling systems are still under development, and existing thermal models are insufficient, although computational fluid dynamics (CFD) simulations provide useful insights into combustion and flow dynamics.

The challenges posed by noise and vibration, which are inherent to the explosive characteristics of PDEs, are significant. Existing literature provides few effective solutions for real-time vibration damping and acoustic suppression suitable for aerospace applications. The integration of PDEs with turbines or fans remains largely undeveloped due to incompatible flow requirements and pressure conditions. Furthermore, the effective combination of PDEs with hybrid systems such as Rotating Detonation Engines (RDEs) has yet to be realized, despite their potential benefits. Additionally, there is a notable deficiency in the advancement of adaptive control systems for PDEs. The lack of real-time monitoring tools and AI-driven fault prediction models obstructs the development of intelligent engines that can adapt to fluctuating operational conditions. Research into scalability and miniaturization is also insufficient, particularly for applications in small UAVs and microsatellites, as most existing PDEs are confined to laboratory-scale models, with minimal efforts to scale them effectively while preserving performance.

Environmental and regulatory issues are another overlooked aspect; the significant noise and high-temperature exhaust produced by PDEs raise questions regarding their adherence to civil aviation standards, yet comprehensive studies on emissions and environmental impacts are scarce. Lastly, the absence of flight-tested PDE systems presents a considerable obstacle, as very few experimental PDEs have advanced beyond ground testing, and long-duration trials are infrequent. Moreover, the lack of investigation into the economic feasibility, mass production methods, and maintenance needs of PDEs further hinders their adoption in commercial and defense sectors. These deficiencies underscore the multidisciplinary challenges confronting PDEs and emphasize the necessity for ongoing research in combustion science, materials engineering, control systems, and aerospace integration. Systematically addressing these issues could significantly propel PDE technology from a theoretical concept to a practical propulsion solution.

5. Objectives Of The Study

- To examine the influence of combustion chamber length on the progression of detonation waves, the efficiency of Deflagration-to-Detonation Transition (DDT), thrust generation, and the overall performance of Pulse Detonation Engines (PDE).
- To identify the ideal ratio of chamber length to diameter that guarantees stable detonation, reduces energy losses, and maximizes space efficiency in compact pulse detonation engine systems.
- To evaluate the influence of fuel injection parameters, such as: Injection timing, Injector configuration, Fuel type (gaseous or liquid), and Mixing characteristics on combustion completeness, ignition reliability, and cycle frequency.
- To examine the impact of sophisticated fuel injection techniques, such as pulsed, swirl, or multipoint injection, on the enhancement of fuel-air mixing and the stabilization of detonation.
- An investigation into the relationship between the length of the combustion chamber and the characteristics of fuel injection, as well as the impact of this interaction on detonation initiation, thermal efficiency, and the reliability of the engine.
- To create simulation models or experimental configurations that effectively depict the interplay between geometric and fuel parameters within authentic PDE operating conditions.
- To analyze the performance compromises associated with varying lengths of combustion chambers in conjunction with diverse injection methods and operational cycle speeds.
- To create design standards and performance metrics that facilitate the advancement of scalable, high-efficiency, and structurally robust PDE systems for aerospace and defense purposes.

6. Methodology

In order to examine the effects of combustion chamber length and fuel injection methods on the performance of a Pulse Detonation Engine (PDE), a modular test engine was designed and constructed. The experimental apparatus consisted of three primary components: a combustion chamber, a detonation chamber, and an exhaust tube. All components were fabricated from Stainless Steel 304 (SS-304) due to its excellent high-temperature strength and resistance to corrosion, which are critical for enduring the extreme thermal and mechanical stresses produced during detonation. The combustion chamber was designed to be interchangeable, featuring three lengths (100 mm, 200 mm, and 300 mm) with a constant internal diameter of 42 mm, thereby allowing for a systematic investigation of the geometric influences on fuel-air mixing and detonation initiation. Fuel delivery was facilitated through a dual-fuel injector system in each combustion chamber, comprising four motorcycle-grade injectors—two for acetylene and two for oxygen. These injectors were arranged circumferentially to ensure uniform mixing within the chamber. The gas flow to each injector was controlled by electronically operated solenoid valves linked to pressurized buffer tanks, guaranteeing a steady and regulated supply of reactants throughout the experiments. This setup allowed for precise management of fuel-to-oxidizer ratios and supported consistent testing conditions.

An Arduino Uno microcontroller controlled the injectors' operation and timing by generating pulse-width modulation (PWM) signals to activate a TIP122 Darlington transistor, which acted as a high-current switch. A $4.76\ \Omega$ resistor limited the current to the injectors, powered by a 12V DC source, allowing precise control over injector activation for a consistent fuel supply. Detonation was initiated by a Shchelkin spiral in a 300 mm detonation chamber, which created turbulence in the fuel-oxidizer mixture to enhance flame acceleration and facilitate deflagration-to-detonation transition (DDT). After ignition, the detonation wave moved through the mixture into a 300 mm exhaust tube, designed for uniform gas flow and stable back-pressure.

To assess the thrust produced during each detonation event, an S-type load cell was incorporated into the engine's thrust stand. This load cell was connected to the National Instruments USB-6501 data acquisition system, which offered 24 digital I/O channels and a 32-bit counter for accurate data collection. The system captured real-time thrust data for every test condition, facilitating a comprehensive analysis of performance trends across various chamber lengths and injection timings. The experimental procedure entailed performing a series of tests for each combustion chamber length while keeping the detonation chamber and exhaust tube conditions constant. The thrust data collected was analyzed to assess detonation efficiency, the impact of combustion chamber geometry, and the influence of injector control on overall engine performance. This approach established a solid framework for understanding the critical factors influencing PDE operation and laid the groundwork for optimizing future designs.

7. Experimental Setup

The experimental setup for the project titled “Investigating the Role of Combustion Chamber Length and Fuel Injection in Pulse Detonation Engine Performance” consists of a modular Pulse Detonation Engine (PDE), which is systematically divided into three primary components: the Combustion Chamber, Detonation Chamber, and Exhaust Tube. The entire system is fabricated using Stainless Steel 340 (SS340), known for its excellent high-temperature resistance and corrosion resistance, making it suitable for repeated detonation cycles and harsh operating conditions.

The performance of the PDE is studied by varying the length of the combustion chamber and analysing the resulting effects on detonation characteristics and engine efficiency.

1. Combustion Chamber

Three interchangeable combustion chambers with lengths of 100 mm, 200 mm, and 300 mm (each with an internal diameter of 52 mm) are employed to investigate the impact of chamber length on fuel-air mixing and detonation initiation. Each chamber is equipped with four motorcycle-grade fuel injectors, mounted circumferentially around the chamber.

- Two injectors are designated for acetylene, and the other two for oxygen.
- The injectors are precisely positioned to ensure efficient mixing and uniform distribution of the fuel-oxidizer mixture prior to ignition.
- The modular nature of the combustion chamber allows for rapid reconfiguration and testing under identical downstream conditions.

2. Detonation Chamber

The detonation chamber has a fixed length of 300 mm and a diameter of 52 mm, identical to that of the combustion chamber to ensure a smooth flow transition. The core feature of this section is a Shchelkin spiral, also 300 mm in length and 52 mm in diameter, installed along the chamber's inner wall.

- The Shchelkin spiral generates flow turbulence and induces flame acceleration, enabling the Deflagration-to-Detonation Transition (DDT).
- The detonation process is initiated and stabilized within this chamber, which is the critical zone for analyzing combustion-to-detonation transformation.

3. Exhaust Tube

Downstream of the detonation chamber is the exhaust tube, which is also 300 mm in length and 52 mm in diameter. This section is responsible for the expulsion of exhaust gases post-detonation.

- The uniform cross-section minimizes flow disturbance and aids in consistent back-pressure development.
- It also serves to safely discharge high-pressure gases and allows complete purging between detonation cycles.

4. Fuel Supply and Control System

- The fuel system utilizes oxygen and acetylene as the oxidizer and fuel, respectively.
- Truck Air Tanks serve as buffer tanks for both gases, maintaining a stable supply.
- The flow from each buffer tank to the respective injector is regulated using electronically controlled solenoid valves.
- This setup ensures precise fuel delivery and reproducible test conditions for each trial.

This configuration allows for controlled experimentation with various combustion chamber lengths, enabling a detailed evaluation of how combustion geometry and injection strategies affect the overall detonation performance of the PDE.



Figure 5 Working Model of Pulse Detonation Engine

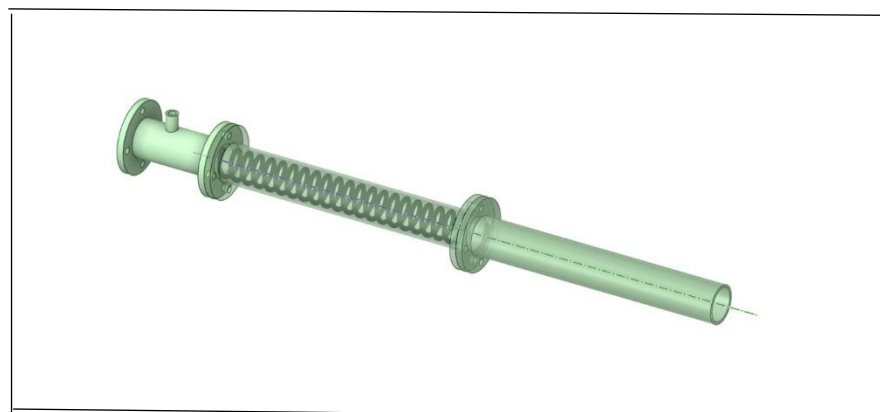


Figure 6 CAD Section View of Shchelkin Spiral and Detonation Chamber



Figure 7 Image of PDE In Action

Fuel Injector Control Using Pulse Modulator

Fuel Injection System

The fuel injection system facilitates the controlled transfer of fuel and oxidizer from the buffer tanks to the combustion chamber. It is designed to ensure safe, reliable, and efficient mixing of reactants while minimizing human intervention. The system is composed of several critical components: fuel injectors, an electronic injection control circuit, solenoid valves, and connecting pipelines.

Fuel Injectors

The experimental configuration employs two motorcycle-grade fuel injectors—one designated for oxygen and the other for acetylene. These injectors are interfaced with the combustion chamber via a Y-shaped connector. This configuration is specifically chosen to prevent premature ignition by ensuring that the fuel and oxidizer do not mix prior to entering the combustion chamber, thus enhancing operational safety and combustion control.

Electronic Injection Control Circuit

An automated electronic circuit is integrated into the injection system to manage the timing and quantity of fuel and oxidizer delivered to the combustion chamber. This automation enhances the precision of injection events and reduces manual oversight. The circuit consists of the following elements:

- **Power Supply:** A 12V DC supply is provided using a Switched Mode Power Supply (SMPS), which converts standard AC input to a regulated DC output required for the circuit's operation.
- **Frequency Modulator:** This component allows for the adjustment of injection frequency and duty cycle. The frequency dictates the number of injection events per unit time, while the duty cycle determines the duration for which the injector remains open during each cycle.

- **MOSFET Switch:** A metal-oxide-semiconductor field-effect transistor (MOSFET) acts as a high-speed, solid-state switch. It receives the Pulse Width Modulated (PWM) signal from the frequency modulator and controls the current supplied to the injectors. The use of a MOSFET ensures rapid switching and protection against back electromotive force (EMF), thereby increasing system reliability.
- **Connection and Control:** Jumper wires are employed to connect the MOSFET to the injectors. The entire circuit is activated or deactivated via an external toggle switch, providing simple and accessible manual control.

Solenoid Valves

A total of four electrically actuated solenoid valves are integrated between the buffer tanks and the injectors—two each for oxygen and acetylene. These valves regulate the flow of gases from the storage tanks and are actuated through electrical signals. Their use enables precise control over gas delivery timing, which is essential for synchronizing with the injection events governed by the electronic control circuit.

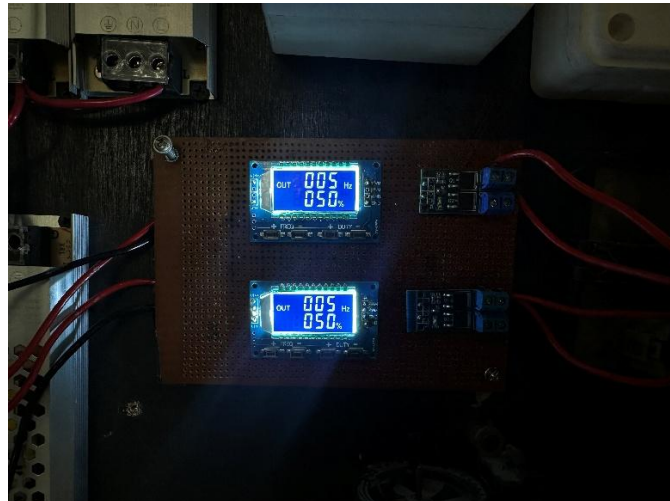


Figure 8 Image of Fuel injector Control Circuit using Pulse Modulator

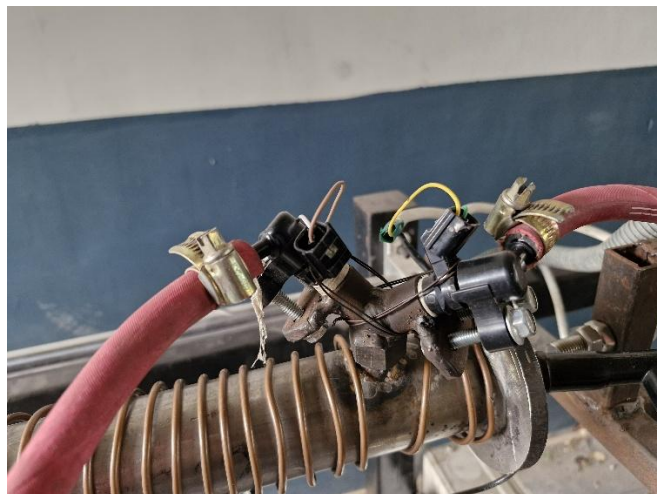


Figure 9 Fuel Injectors with Y Section



Figure 10 Buffer Tank



Figure 11 Solenoid Valve



Figure 12 Spark Plug

Alternate Circuit for Fuel Injector Control Circuit Using Arduino UNO

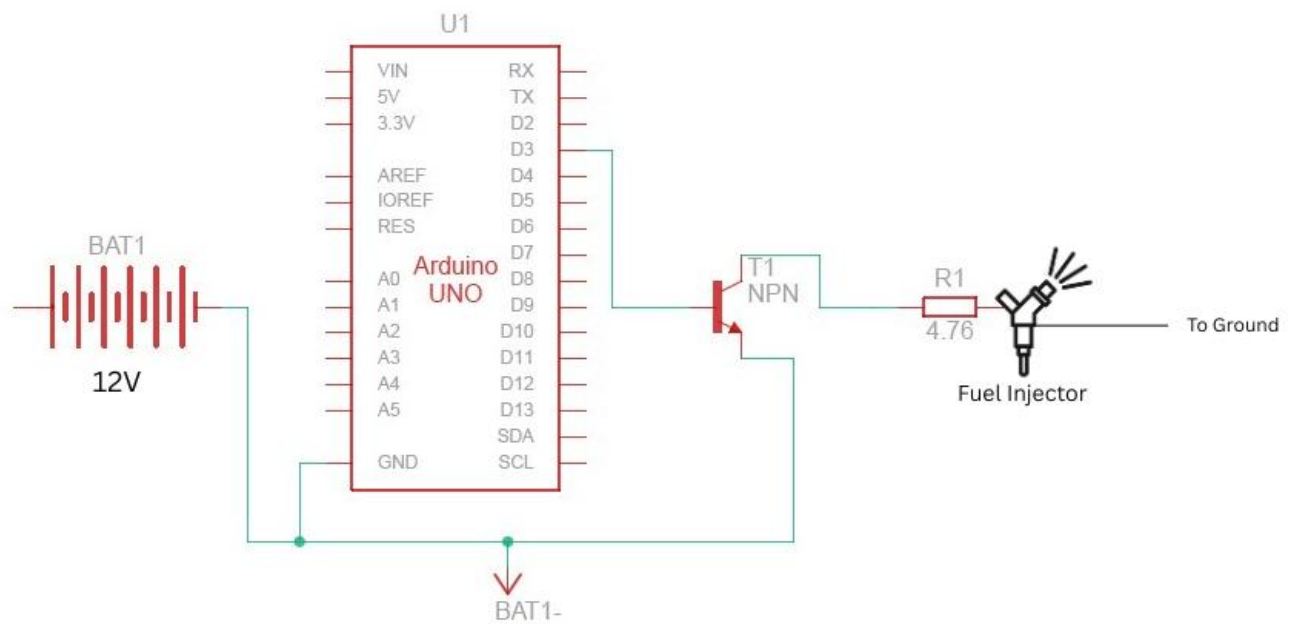


Figure 13 Schematic Diagram of Injector Control Circuit

Arduino UNO

The Arduino Uno is a widely used microcontroller platform based on the ATmega328P microcontroller. It is equipped with 14 digital input/output pins, six of which can generate pulse-width modulation (PWM) signals, making it suitable for precise control applications such as fuel injection. With its open-source architecture, ease of programming using the Arduino Integrated Development Environment (IDE), and robust hardware design, the Arduino Uno serves as the central controller for generating timed pulses to drive the fuel injectors.

The proposed circuit design utilizes the Arduino Uno to produce a PWM signal that switches the TIP122 transistor. The TIP122 acts as a high-current switch, enabling the control of the injector's operation.

Components Used

1. **Arduino Uno:** Acts as the controller to generate the necessary control signals for the fuel injectors.
2. **Fuel Injector:** An electromechanical device to inject fuel into the combustion chamber.
3. **TIP122 Transistor:** A Darlington transistor used as a switch to control the high current required by the fuel injectors.

4. **4.76 Ω Resistor:** Used to limit current flow to the injector, ensuring it operates within safe limits.
5. **12V Power Supply:** Supplies the required voltage and current for the injectors and the Arduino.
6. **Connecting Wires:** For interfacing all components.
7. **Breadboard or PCB:** For assembling the circuit.

Circuit Design

1. Power Source:

- The fuel injectors typically operate at 12V DC. A separate regulated power supply is recommended to prevent electrical noise from affecting the Arduino.

2. Arduino Output:

- A digital output pin (e.g., pin 9) is configured to generate control pulses to the TIP122 transistor. The duty cycle of the pulses determines the injector's ON time.

3. Switching Circuit:

- The base of the TIP122 transistor is connected to the Arduino output, ensuring reliable switching.
- The emitter of the TIP122 is connected to ground.
- The collector of the TIP122 is connected to one terminal of the 4.76 Ω resistor.

4. Fuel Injector:

- The positive terminal of the fuel injector is connected to the 12V power supply.
- The negative terminal is connected to the free end of the 4.76 Ω resistor.

Working Principle

The working principle of the fuel injector control circuit revolves around precise management of the injector's ON and OFF states using a PWM signal generated by the Arduino Uno. The PWM signal, outputted from a designated pin, controls the TIP122 transistor, which acts as a high-current switch. A 4.76 Ω resistor is incorporated into the circuit to regulate the current flowing through the fuel injector, ensuring safe operation. The current can be calculated using Ohm's law ($I = V/R$), where V is the voltage across the resistor. When the transistor is in the ON state, current flows through the injector, causing it to open and inject fuel. Conversely, when the transistor is OFF, the current flow is interrupted, and the injector closes, halting fuel delivery. This precise control mechanism

ensures effective and reliable operation of the fuel injector in synchronization with the PDE's requirements.

Arduino Code

```
// C++ code
//
// Define pin
// Define pin
int injectorPin = 3; // Connect to the TIP 122 base
void setup() {
    pinMode(injectorPin, OUTPUT); // Set pin as output
}
void loop() {
    digitalWrite(injectorPin, HIGH); // Turn injector ON
    delay(1000);                      // Wait for 1 second
    digitalWrite(injectorPin, LOW);  // Turn injector OFF
    delay(1000);                      // Wait for 1 second
}
```

The delay can be adjusted according to the frequency requirements of the PDE

Data Acquisition System

The National Instruments USB-6501 (779205-01) data acquisition system is an essential element of the experimental framework for the pulse detonation engine (PDE) project. This compact, bus-powered USB digital I/O system is engineered to deliver dependable and high-precision data acquisition capabilities. It features 24 digital I/O channels, comprising 8 inputs, 8 outputs, and 8 bidirectional lines, providing considerable flexibility to meet various experimental needs. Moreover, the USB-6501 is equipped with a 32-bit counter, which is particularly advantageous for applications that demand precise timing and counting functionalities. With an operational voltage range of -0.5 V to $+5.8\text{ V}$, the system is designed with robust overvoltage protection, ensuring its resilience in challenging environments. Its USB 2.0 interface facilitates seamless connectivity with computing platforms, allowing for easy integration into a variety of experimental setups. Additionally,

its compact design enhances portability, making it suitable for both laboratory and field applications. In this research, the USB-6501 is employed to connect with an S-type load cell, a vital component for measuring the thrust generated by the PDE. S-type load cells are well-regarded for their exceptional accuracy and reliability in force measurement applications. Their distinctive shape not only contributes to their structural integrity but also guarantees uniform stress distribution under load. These load cells can measure both tensile and compressive forces, rendering them highly adaptable for experimental investigations involving dynamic forces. The load cell transforms mechanical force into a corresponding electrical signal, which is then captured and processed by the USB-6501.

The combination of the S-type load cell and the USB-6501 establishes the core of the data acquisition system, allowing for accurate and instantaneous monitoring of the thrust produced by the PDE. The load cell's digital outputs are sampled and documented, enabling comprehensive analysis of engine performance across different conditions. The capabilities of the USB-6501 guarantee that essential data is captured with exceptional precision, rendering it an essential instrument for evaluating the effectiveness of design alterations in the PDE, including modifications to the combustion chamber length and the addition of fuel injectors to enhance atomization. The harmonious interaction between the load cell and the data acquisition system offers a dependable framework for validating theoretical models and refining engine performance.



Figure 14 NI DAQ



Figure 15 S-type load cell

8. Result

Thrust Readings at Varying Frequency

Combustion Chamber Length	Frequency		
	1Hz	5Hz	10Hz
100mm			
200mm			
300mm			

9. Conclusion

Detonation-based propulsion systems, such as Pulse Detonation Engines (PDEs) and Rotating Detonation Engines (RDEs), signify a significant advancement in aerospace technology, offering the potential for greater efficiency and improved performance relative to conventional engines. Although PDEs are notable for their straightforward design and versatility across different flight conditions, their challenges related to stability and mechanical durability limit their broader implementation. In contrast, RDEs, which utilize a continuous detonation process, present a more stable and efficient option; however, they encounter difficulties associated with wave dynamics, material fatigue, and compatibility with current aerospace systems. This research highlights the necessity for additional investigation into advanced materials, optimized combustion chamber configurations, and enhanced fuel injection mechanisms to address these challenges. By addressing these issues, detonation engines could transform propulsion technologies, facilitating quicker, more efficient, and environmentally sustainable aerospace operations.

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